CN Assignment # 2

M. Baqar Laghari
24K-0006
BAI-5A

Task 1:

1. Casualty lists, supply requests and rescue assignments must reach right team correct and complete. A lost or corrupt packet could send a team to wrong place and delay help. Reliable transfer detects errors and recovers lost data and deliver everything in order.

2. A 1000 byte packet takes only 4 ms to send at 2 Mbps but the RTT is 500 ms. Stop and Wait sends one packet, then sits idle waiting for it's ACK. So the link is used only about 0.794% of the time.

3. Sender can help sending several packets while earlier ACKs are still travelling. This fills long satellite waiting time with useful work. With N=4; utilization is about 3.175%, four times higher than Stop and Wait.

CamScanner

4. SR is more suitable, for two reasons:

- It retransmits only lost packet, so low of limited bandwidth is wasted than with GBN resending whole windows.
- The reciever buffers out of order packets and delivers them in order once the gap is filled, so correct packets are never thrown away.

Cost is more reciever money and per-packets timers.

Task 2:

Assumptions:

- Sender sends whenever window has space.
- ACKs 0 and 1 before packet 2 times out, so packets 4 and 5 get sent.
- No other packet or ACK is lost.
- Timeout is long enough for SR sender to recieve ACKs 3, 4 and 5 before packet 2 times out.
- Windows are clipped at 9.
- In GBN, ACK n means "everything up to n recieved." In SR, ACK n means "only n recieved."

CamScanner

(a) Go-Back-N (N=4)

#	Sender	PKT	Reciever	ACK	Sender window
1	Send 0	0	Deliver, expected=1	ACK0	[0-3]
2	Send 1	1	Deliver, expected=2	ACK1	[0-3]
3	Send 2	2	Lost	-	[0-3]
4	Send 3	3	Out of order, discard expected=2	ACK 1 (duplicate)	[0-3]
5	Recieve ACK0, Send 4	4	Discard, expected=2	ACK 1 (dup)	[1-4]
6	Recieve ACK 1, Send 5	5	Discard, expected=2	ACK 1 (dup)	[2-5]
7	Ignore 3 dup ACK 1s	-	-	-	[2-5]
8	Timeout for base=2, resend 2,3,4,5	2-5	-	-	[2-5]
9	Wait	2,3,4,5	Deliver all in order, expected=6	ACK 2,3,4,5	[2-5]
10	Recieve ACK2 → send 6	6	In flight	ACK 2 arrives	[3-6]

CamScanner

#	Sender	PKT	Reciever	ACK	sender window
11	Recieve ACK 3 → send 7	7	In flight	ACK3 arrives	[4-7]
12	Recieve ACK 4 → send 8	8	In flight	ACK4 arrives	[5-8]
13	Recieve ACK 5 → send 9	9	In flight	ACK5 arrives	[6-9]
14	Wait	6-9	Deliver all expected = 10	ACK 6,7,8,9	[6-9]
15	Recieve ACK 6,7,8,9	-	Done	-	[7-9] → [8-9] → [9] → empty.

GBN Transmission total: 10 (original) + 4 (retransmission) = 14 transmission (2, 3, 4, 5)

(b) Selective Repeat (N=4 at both sides)

#	Sender	PKT	Reciever	ACK	S window / R window
1	Send 0	0	In flight	-	[0-3] / [0-3]
2	Send 1	1	In flight	-	[0-3] / [0-3]
3	Send 2	2	Loss	-	[0-3] / [0-3]
4	Send 3	3	In flight	-	[0-3] / [0-3]
5	Wait	0	Deliver=0 rcv base=1	ACK 0	[0-3] / [1-4]
6	Wait	1	Deliver=1 rcv base=2	ACK 1	[0-3] / [2-5]
7	Wait	3	Buffer 3, no delivery	ACK 3	[0-3] / [2-5]
8	Reciere ALK0 Send 4	4	Buffer [3,4]	ACK 4	[1-4] / [2-5]
9	Reciere ACK 1 Send 5	5	Buffer [3,4,5]	ACK 5	[2-5] / [2-5]
10	Timeout, resend dy 2	-	Buffer [3,4,5]	-	[2-5] / [2-5]

CamScanner

#	Sender	PKt	Reciever	ACK	[S window] / [R window]
11	Timeout, resend only 2	2	In flight	–	[2-5] / [2-5]
12	Wait	2	Deliver 2,3,4,5 buffer empty	ACK 2	[2-5] / [6-9]
3	Recieve ACK2 base=6, send 6,7,8,9	6-9	In flight	ACK 2 arrives	[6-9] / [6-9]
4	Wait	6,7, 8,9	Deliver each in turn, rcv base = 10	ACK 6,7,8,9	[6-9] / empty
5	Recieve ACK 6,7,8,9	–	Done	–	[7-9] → [8-9] → [9] → empty

SR Total: 10 (original) + 1 (retransmission) (2) = 11 transmissions.

GBN reciever throws away packets 3,4 and 5 because they arrive out of order. So after the timeout sender must resend 2, 3, 4 and 5, which is 4 retransmission. While in SR, reciever buffers 3, 4, 5 and ACK each one individually, so only one packet is resent, which is 1 retransmission.

CamScanner

Task 5:

(1) Given: $L = 8000$ bits

$R = 2 \times 10^6$ bps

$RTT = 0.5$ ms

$L/R = \frac{8000}{2{,}000{,}000} = 0.004\text{s} = 4\text{ms}$

Stop-and-Wait:

$U = L/R \;/\; (RTT + L/R)$

$= \frac{0.004}{(0.5 + 0.004)} = 0.007936 = 0.794\%$

Pipelining, N=4:

$U = \frac{N(L/R)}{(RTT + L/R)}$

$= \frac{4(0.004)}{5 + 0.004} = 0.031746 = 3.175\%$

Pipelining lets sender send 4 packets in time Stop and Wait sends 1, so utilization improves 4 times. RTT is still much larger than 4 ms transmission time, so utilization stays low. These are ideal, no error values.

CamScanner

(2)

(a) $2^3 = 8$ sequence numbers, 0, 1, 2, 3, 4, 5, 6, 7.

(b) SR needs $N \leq \frac{M}{2}$, so $N \leq \frac{8}{2} = 4$.

Minimum safe window is 4.

(c) Take window $N = 5$.

→ Sender sends 0-4 but all 5 ACKs are lost while reciever accepts remaining.

→ Reciever's new window becomes {5, 6, 7, 0, 1}

→ Sender times out and resends old packages.

→ Number 0 is inside reciever's new window, so reciever buffers it as a new packet 0. This is ambiguity.

→ Later, old data could be wrongly delivered as new data.

Initial window

0	1	2	3	4

New window

5	6	7	0	1

CamScanner